

Master Math for JEE Main & JEE Advanced

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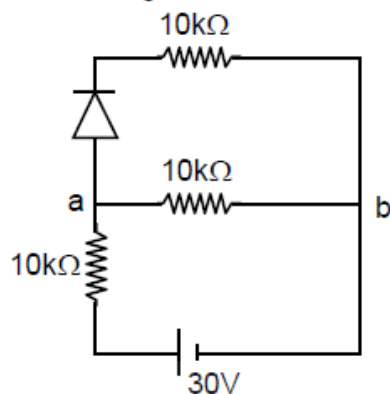
JEE Mains 2020 Jan Chapter wise Question Bank

Semiconductors

Q1

There is a electric circuit as shown in the figure. Find potential difference between points a and b.

चित्रानुसार परिपथ में एक विद्युत धारा है, तो बिन्दुओं a तथा b के मध्य विभवान्तर ज्ञात करो।



(1) 0V

(2) 15V

(3) 10V

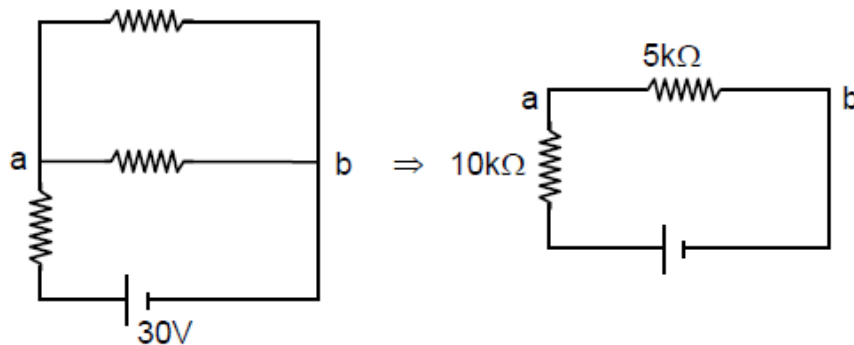
(4) 5V

Sol

(3)

Diode is in forward bias, so it will behave as simple wire so,

डायोड अग्र बायस अवस्था में है, अतः यह साधारण तार की तरह कार्य करेगा।

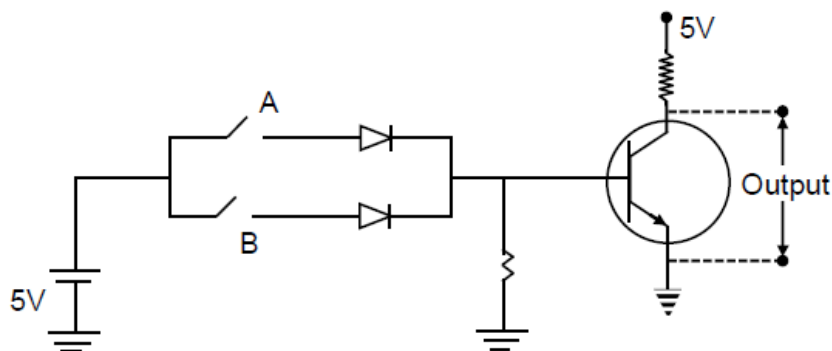


So अतः, $V_{ab} = \frac{30}{5+10} \times 5 = 10V$

Q2

Choose the correct Boolean expression for the given circuit diagram:

निचे दर्शाये गये परिपथ के लिए बूलियन व्यंजक ज्ञात करो।



(1) $A \cdot B$

(2) $\bar{A} + \bar{B}$

(3) $A + B$

(4) $\bar{A} \cdot \bar{B}$

8th Jan Morning

Sol

(4)

First part of figure shown is OR gate and

आरेख का प्रथम भाग OR द्वार प्रदर्शित करता है

Second part of figure shown is NOT gate

तथा द्वितीय भाग NOT द्वार प्रदर्शित करता है।

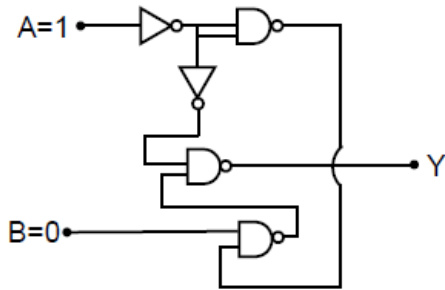
So अतः $Y_p = \text{OR} + \text{NOT} = \text{NOR gate}$

$$Y = \overline{A + B} = \bar{A} \cdot \bar{B}$$

Q3

Output at terminal Y of given logic circuit.

दिये गये तार्किक परिपथ के लिये, सिरे Y पर निर्गत मान है।



- (1) 1
 (2) 0
 (3) Not determine
 ज्ञात नहीं कर सकते
 (4) Oscillating between 0 and 1
 0 तथा 1 के बीच में दोलन करेगा।

8th Jan Evening

Sol

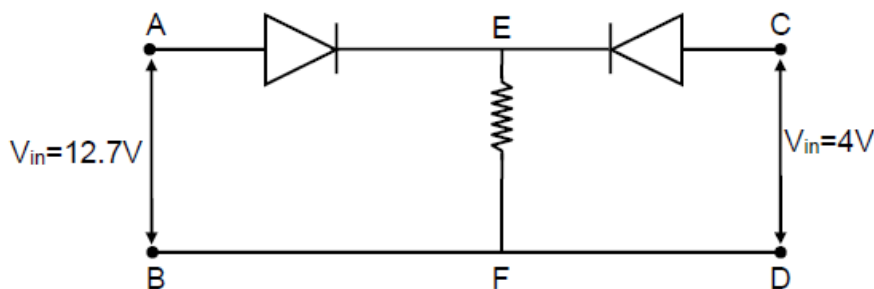
(2)

$$\begin{aligned}
 Y &= \overline{AB} \cdot A \\
 &= \overline{AB} + \overline{A} \\
 &= AB + \overline{A} \\
 &= 0 + 0 \\
 &= 0
 \end{aligned}$$

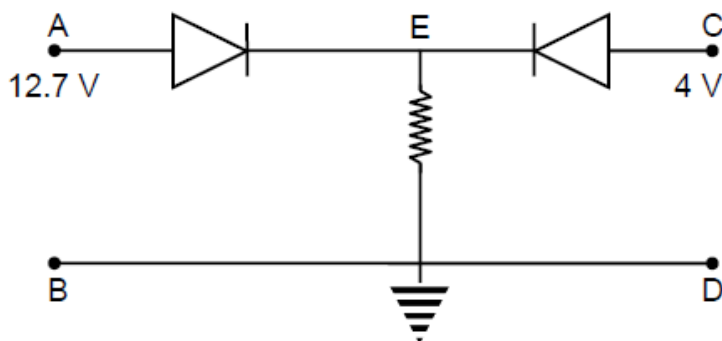
Q4

In the given circuit both diodes having zero forward resistance and built-in potential of 0.7 V. Find the potential of point E in volts.

दिये गये परिपथ में दो डायोड, जिनका अग्र अभिनती प्रतिरोध शून्य है और अवरोधी विभव 0.7 V है तब बिन्दु E का विभव वोल्ट में होगा।

9th Jan Morning

12



Let $V_B = 0$

Right diode is reversed biased and left diode is forward biased

माना $V_B = 0$

दायां डायोड पश्च अभिनती तथा बायां डायोड अभिनती में है।

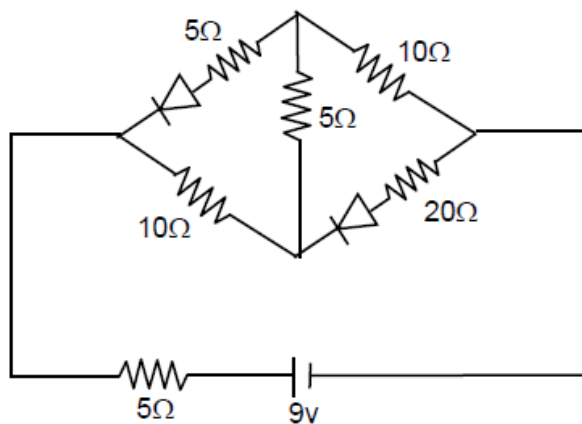
$$\therefore V_E = 12.7 - 0.7$$

$$= 12 \text{ Volt}$$

Q5

Find the current supplied by the battery?

परिपथ में बहने वाली धारा ज्ञात करो?



(1) 0.1 A

(2) 0.3 A

(3) 0.4 A

(4) 0.5 A

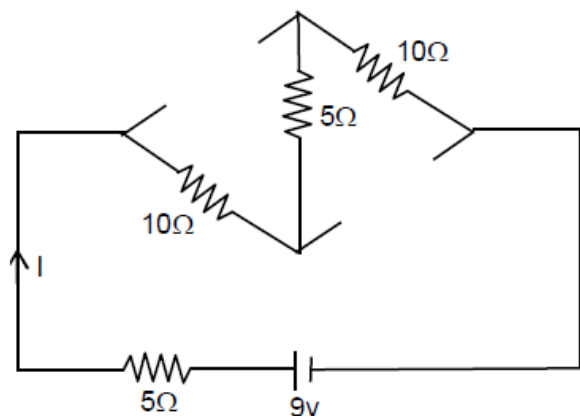
9th Jan Evening

Sol

(2)

Both diodes are in reverse biased

दोनों डायोड पश्च बायस स्थिति में है।

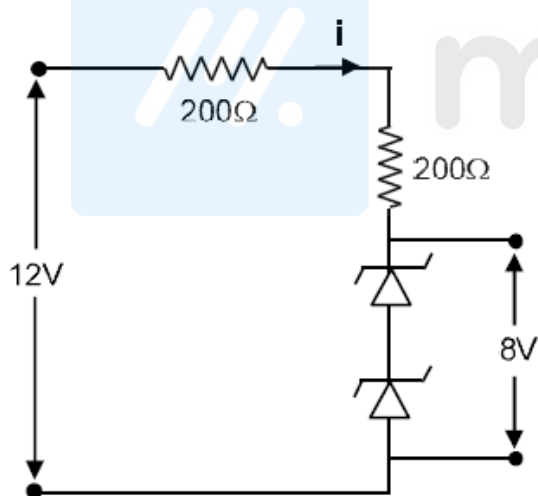


$$I = \frac{9}{30} = \frac{3}{10} \text{ A} = 0.3 \text{ A}$$

Q6

Find the power loss in each diode (in mW), if potential drop across the zener diode is 8V.

यदि प्रत्येक जिनर डायोड का विभव 4V है तो प्रत्येक डायोड में शक्ति हानि (mW में) ज्ञात करो—

9th Jan Evening

Sol

40

$$i = \frac{(12 - 8)}{(200 + 200)} \text{ A} = \frac{4}{400} = 10^{-2} \text{ A}$$

Power loss in each diode प्रत्येक डायोड में शक्ति हानि = $(4)(10^{-2}) \text{ W} = 40 \text{ mW}$

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